Block by Bio-Block

Applying Bioplastics for LEGO-like Building Block Production

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Humanity's increasing dependence on plastic is apparent in many aspects of our daily lives, including the world of children's toys. As kids lose interest and outgrow their toys, they often end up in landfills, where they remain a substantial environmental burden. In the United States alone, the toy industry accounts for over three billion tons of plastic used, according to The World Counts ("Environmental Impact of Toys," n.d.). In this research, we will strive to analyze the timeless children's toy, the LEGO, for its physical and chemical properties to propose an eco-conscious solution using bioplastics. Bioplastics, derived from natural polymers such as starch or cellulose, provide a promising alternative to curb the environmental impact of plastic production.

LEGOs are made from ABS plastic, short for Acrylonitrile Butadiene Styrene. The chemical formula of Acrylonitrile Butadiene Styrene is (C8H8·C4H6·C3H3N)n. The name composition describes the three types of subunits, or branches, which each contribute to the polymer's overall physical and chemical properties. Firstly, acrylonitrile has a vinyl group attached to a nitrile group, contributing to the chemical resistance & heat stability of ABS. Secondly, 1,3-butadiene is a butene molecule where the end carbons are double-bonded, delivering hardness & impact strength. Lastly, styrene consists of a vinyl group as a substituent on a benzene, providing rigidity & processability. Linking these three molecules together in a polymer chain (as depicted in diagram 1) creates ABS plastic, characterized for its high rigidity, high impact resistance, dimensional stability, and a bright surface.

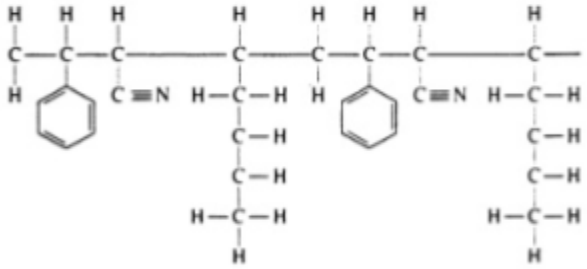


DIAGRAM 1. Chemical structure of the ABS polymer

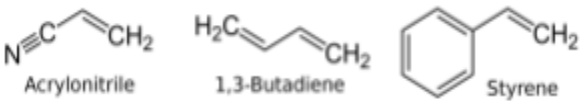


DIAGRAM 2. Chemical structure of branches

Producing ABS plastic involves emulsion or continuous mass polymerization. The three monomers–acrylonitrile, butadiene, and styrene–are combined in specific ratios and subjected to a polymerization reaction. To initiate the chemical reaction. Heat and agitation are applied. As the reaction progresses, the heterogeneous mixture undergoes stabilization, and the polymer chains multiply. After polymerization is complete, the ABS congeals, separating the polymer particles from the water. Further

steps, such as washing and drying, can remove impurities and residual chemicals. After processing, ABS is usually stored as pellets, which improves the ease of manufacturing processes like injection molding. The uncoloured or natural off-white pellets can then be coloured using vibrant dyes during later stages of the manufacturing process.

ABS plastic binds well to dyes, allowing for the vibrant colours we see in LEGO bricks. However, that is simply one of the reasons LEGOs are made from ABS plastic. It was also chosen for its low cost, versatility, and durability. LEGO sets are often incredibly complex, with some sets having hundreds of unique pieces, which complicates the manufacturing process. Low cost and versatility of the materials are thus incredibly important as one material must produce countless different parts while maintaining relative affordability. Durability is also crucial as LEGO blocks must withstand many years of play, being built and rebuilt countless times over. In addition, the moldability and low melting temperature of ABS plastic make it suitable for injection molding the LEGO bricks. The material choice and precision in manufacturing is what allows for the signature “click” noise two blocks make when building. For designing our LEGOS, there are three signature characteristics we will strive to reproduce: moldability, stiffness when hardened, and functionality for building (creating the signature “click”).

Having learned more about ABS plastics and their reason for having been chosen, we can begin to look at bioplastic alternatives that fit this criteria. There are various types of bioplastics, each with distinct differences and benefits to be compared. Firstly, starch-based plastics, as predicted by the name, are made from cornstarch or cassava. They are versatile and moldable, but limited in terms of moisture sensitivity, mechanical strength, and durability. Next, protein-based plastics can be made from wheat gluten or casein in milk. They have many of the same properties as starch-based bioplastics, including Lastly, cellulose-based bioplastics are made from cellulose fibres of plants. They exhibit great strength and durability, but may have higher production costs. Due to its superior strength, cellulose-based bioplastics will likely work the best. Since we are producing the LEGOs on a much smaller scale, cost will not hinder our production. For the sake of this experiment, we will test multiple types of bioplastics to see which best fits the criteria of moldability, stiffness when hardened, and functionality for building.

Pre-lab Questions

What is a polymer?
A polymer is a material composed of long chains of molecules. Polymers are long-chain molecular structures found in plastics. They consist of large molecules called macromolecules that are made up of many repeating units called monomers. This is further categorized into natural polymers like protein and cellulose versus synthetic polymers like polyethylene.

Why are scientists exploring alternatives to petroleum-based plastics?
Scientists are looking into alternatives to petroleum-based polymers because of environmental concerns. Traditional plastics derived from petroleum are not biodegradable and can linger in the environment for hundreds of years, producing pollution. Bioplastics are a more sustainable alternative since they are frequently made from renewable biomass sources and may degrade naturally.

What does biodegradable mean?
Biodegradability is the ability of a substance to degrade spontaneously over time, usually through the activities of bacteria or fungi. Biodegradable polymers can break down into simpler, more ecologically friendly materials.

What are two advantages of bioplastics over biodegradable plastic and petroleum-based plastics?
i. Bioplastics are frequently made from renewable biomass sources, making them more sustainable.
ii. Many bioplastics are biodegradable, which means they can be broken down into CO2 and water by microbes.

What are biopolymers?
Biopolymers are polymers (long chains of molecules) formed from biological sources like plants and animals. They are created to be and are found to be biocompatible and biodegradable. In bioplastics, biopolymers are the primary components derived from renewable biomass sources such as fermented sucrose, gelatin, vegetable oil, maize starch, pea starch or cellulose.

Objective

To compare and determine the physical properties of starch-based and cellulose-based plastics, and to test their potential as eco-friendly alternatives to traditional petroleum-based plastics.

Materials

Starch-based Bioplastic Ingredients:

15g of Cornstarch
2ml of Glycerol
10ml of Vinegar
100ml of Water
Food colouring
3D-printed LEGO mold
2x - 250ml Beaker
Stir Stick
Hot Plate

Cellulose-based Bioplastic Ingredients:

15 g of Cellulose fibres (extracted from plant sources)
2ml of Glycerol
100ml of Water
Food colouring (as desired)
3D-printed LEGO mold
1x - 250ml Beaker
Stir Stick

Cellulose-based Bioplastic Ingredients (No Glycerol):

15 g of Cellulose fibres (extracted from plant sources)
100ml of Water
Food colouring (as desired)
3D-printed LEGO mold
1 - 250ml Beaker
Stir Stick

Cellulose-based Bioplastic Ingredients (No Water) :

15 g of Cellulose fibres (extracted from plant sources)
5ml of Glycerol
Food colouring (as desired)
3D-printed LEGO mold
1 - 250ml Beaker
Stir Stick

For all: Weights for compression and tensile strength tests

Procedure

<u>Bioplastic Synthesis:</u>

a. Starch-based Bioplastic:

1. Mix 15g of cornstarch and 100 ml of water in a beaker.
2. Mix 10 ml of vinegar and 2 ml of glycerol in a separate beaker.
3. Combine the two mixtures and stir well until a uniform consistency.
4. Add food colouring if desired.
5. Heat the mixture on a hot plate, stirring continuously until it thickens and becomes moldable.

b. Cellulose-based Bioplastic:

1. Mix 15g of cellulose fibres and 100ml of water in a bowl until uniform consistency.
2. Add 2 ml of glycerol and mix.
3. Stir the mixture thoroughly until a gel-like consistency is achieved.
4. Add food colouring if desired.

c. Cellulose-based Bioplastic (No Glycerol):

1. Mix 15g of cellulose fibres and 100ml of water in a bowl until uniform consistency.
2. Stir the mixture thoroughly until a gel-like consistency is achieved.
3. Add food colouring if desired.

d. Cellulose-based Bioplastic (No Water):

1. Measure 15g of cellulose fibres
2. Add 5 ml of glycerol and mix.
3. Stir the mixture thoroughly until a putty-like consistency slowly add glycerol until the consistency is achieved.
4. Add food colouring if desired.

<u>Molding Process:</u>

a. Starch-based Bioplastic:

1. Pour the heated starch-based bioplastic into the 3D-printed LEGO mold.
2. Close the mold to shape the bioplastic into a LEGO brick.
3. Allow the bioplastic to cool and solidify in the mold.

NOTE: overfill the mold slightly so the excess material can be pushed out to limit air pockets in the mold. Additionally, read testing section C for additional steps.

b. Cellulose-based Bioplastic:

1. Pour the cellulose-based bioplastic into the 3D-printed LEGO mold.
2. Close the mold to shape the bioplastic into a LEGO brick.
3. Allow the bioplastic to cool and solidify in the mold.

NOTE: be sure to overfill the mold slightly so the excess material can be pushed out to limit air pockets in the mold. Additionally, read testing section C for additional steps.

c. Cellulose-based Bioplastic (No Glycerol):

1. Pour the cellulose-based bioplastic into the 3D-printed LEGO mold.
2. Close the mold to shape the bioplastic into a LEGO brick.
3. Allow the bioplastic to cool and solidify in the mold.

NOTE: be sure to overfill the mold slightly so the excess material can be pushed out to limit air pockets in the mold. Additionally, read testing section C for additional steps.

d. Cellulose-based Bioplastic (No Water):

1. Pour the cellulose-based bioplastic into the 3D-printed LEGO mold.
2. Close the mold to shape the bioplastic into a LEGO brick.
3. Allow the bioplastic to cool and solidify in the mold.

NOTE: be sure to overfill the mold slightly so the excess material can be pushed out to limit air pockets in the mold. Additionally, read testing section C for additional steps.

Testing:

a. Stiffness Test:

1. Place weights on top of each bioplastic LEGO brick to test its compression strength.
2. Suspend each brick on its side and attach weights to check its tensile strength.

b. Compatibility Test:

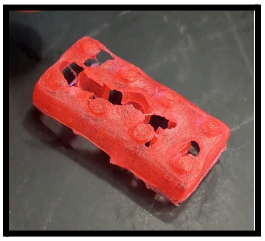
1. Attempt to connect each bioplastic LEGO brick with commercially available ABS plastic LEGO bricks.
2. Observe and record the compatibility of each bioplastic with standard LEGO bricks.

c. Moldability Test:

1. Repeat the same process with the cellulose-based bioplastic LEGO brick.
2. Assess the moldability by checking how well the bioplastic conforms to the intricate details of the mold.
3. Record any deviations from the original LEGO mold dimensions.
4. Record observations on the ease of molding.

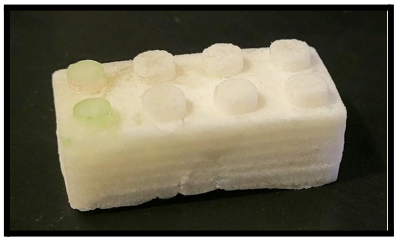
Note: Repeat the entire procedure for both types of bioplastics to ensure accurate comparisons. Record observations, measurements, and any notable differences between the starch-based and cellulose-based bioplastics.

Observations



Cellulose + water

Cellulose glycerin & water



Test Data

	Moldability	**Stiffness**	**Compatibility**
Starch glycerin	High Moldability High Shrinkage	Medium Long drying time	Low compatibility due to high shrinkage Ideal plastic feel and texture
Cellulose glycerin & water	Medium - High Medium Shrinkage	Medium-High Medium drying time	Medium-low compatibility due to shrinkage Ideal plastic feel and texture
Cellulose just water	Medium - High Medium Shrinkage	Medium-High Medium drying time	Medium-low compatibility due to shrinkage Ideal plastic feel and texture
Cellulose just glycerin	Low Moldability Low Shrinkage	High Short drying time	High compatibility Performs like Lego with the greatest stiffness. Unique feel and look

Photos of the Lego blocks (cellulose with only glycerin)

Results

The starch bioplastic came out the least compatible with a commercial Lego. The liquid mixture exhibited a gel-like structure that did not reach the ideal texture and viscosity that was looked for when wanting to begin the molding process. The drying process was the longest out of all of the molds taking about three to four days to finally solidify into a plastic. Similar to the other mixtures, the starch exhibited major shrinkage to its overall shape. Additionally, the type of plastic created was a more malleable plastic suitable for a task that needed to bend while also keeping shape such as potentially for a straw. This did not match the Lego structure that was looked for and matches the functionality the least. A change in the proportions would additionally benefit the outcome by limiting the glycerol to make a tougher plastic while also limiting the water.

Cellulose with just water and cellulose with water and glycerol came out fairly identical for all steps until the drying process. Both mixtures exhibited a gel-like structure that had easy moldability that shaped the mold perfectly. The drying and plastification time between the two substances is essentially the same, with both drying at similar speeds. Though the moldability was easy the mixtures lost mass and shape due to shrinkage with the cellulose with glycerin and water shrinking slightly more than its water-only counterpart. The functionality, in turn, was limited as the result was way too thin to be utilized as a Lego, this is also caused by the mold shape and size. However, the potential for these materials is bright as they most closely resemble commercial Lego pieces while maintaining strength and durability.

Cellulose with just glycerol made a tough putty-like substance, best described as stiff playdough, which would not mold to full Lego mold. The overall drying time for the material was the shortest out of all of the compositions only needing to wait a few hours for it to get hard as the putty itself was already tough. It was compressed into the top half of the Lego mold and has the least shrinkage and greatest functionality as it can be used to connect actual Lego pieces to it. This mixture overall fits the best functioning piece from our testing. Further testing can create a more suitable bioplastic Lego alternative.

The following are ranked in terms of most to least overall effectiveness: glycerol and water, cellulose with just water, cellulose with just glycerol and starch coming in at last due to drastic shrinkage. Starch, due to its drastic shrinkage and flimsy composition, is more usable for alternative items that require malleable plastic.

Conclusion

The lab was ultimately partially successful in creating a compatible LEGO block. For future experiments, various adjustments can be made based on the observations to improve the structure, strength, and compatibility. Major adjustments would include the adjustment in the mold to make it more accountable for shrinkage and material. The implication of thicker walls and a thicker frame for the Lego allows for the material to dry more efficiently while also keeping its functionality. An alternative mold would remove the air trapped when moulding the plastic. It essentially keeps the top and bottom connective parts closed while filling the rest with bioplastic rather than being hollow.

Next comes adjustments in measurements, mainly the change in the starch proportions. To create a more gel-like structure in the mixture the proportions of starch, vinegar and water need to be changed. A more gel-like structure requires less water in the overall reaction. The gel-like structure begins to form as the addition of heat is introduced to the substance. In the original procedure this occurred however was watered down by the extra water in the overall mixture. The proportions of the starch mixture would likely change to one along the lines of 50g of cornstarch, 2ml of glycerol, 15ml of vinegar and 35ml of water. This allows for a closer ratio of equal parts water and starch. The process creates a higher viscosity before the introduction of glycerol and heat overall allowing for easier formation of a gel-like structure. This further would lessen the shrinkage as it limits the amount of oxygen the bioplastic needs to let out.

Lastly, a future experiment would involve a solution to prevent shrinkage for both the cellulose with just water and the cellulose with glycerin and water. Layering the plastic in the mold with multiple runs of the mixture would fully fill the mold and give it time to dry consistently. This allows for the material to seep into the holes made by the shrinkage of the dried portions and act as a glue to add more thickness to the overall Lego. Overall, adding the mixture multiple times to the same mold will allow the gaps between the product to be filled and dry together.

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